

Human Genetics and Genomics for Drug Target Identification and Prioritization: Open Targets' Perspective

Ellen M. McDonagh,^{1,2,3} Gosia Trynka,^{1,3}
Mark McCarthy,⁴ Emily Rose Holzinger,⁵
Shameer Khader,⁶ Nikolina Nakic,⁷ Xinli Hu,⁸
Helena Cornu,^{1,2} Ian Dunham,^{1,2,3}
and David Hulcoop^{1,2,3}

¹Open Targets, Wellcome Genome Campus, Hinxton, UK; email: emcdonagh@ebi.ac.uk

²European Molecular Biology Laboratory, European Bioinformatics Institute, Wellcome Genome Campus, Hinxton, UK

³Wellcome Sanger Institute, Wellcome Genome Campus, Hinxton, UK

⁴Genentech, South San Francisco, California, USA

⁵Bristol Myers Squibb, Cambridge, Massachusetts, USA

⁶Precision Medicine & Computational Biology, Sanofi, Cambridge, Massachusetts, USA

⁷Genomic Sciences, GSK, Stevenage, UK

⁸Inflammation and Immunology, Pfizer Research and Development, Inc., Cambridge, Massachusetts, USA

ANNUAL REVIEWS CONNECT

www.annualreviews.org

- Download figures
- Navigate cited references
- Keyword search
- Explore related articles
- Share via email or social media

Annu. Rev. Biomed. Data Sci. 2024. 7:59–81

First published as a Review in Advance on
April 12, 2024

The *Annual Review of Biomedical Data Science* is
online at biomedata.annualreviews.org

<https://doi.org/10.1146/annurev-biomedata-102523-103838>

Copyright © 2024 by the author(s). This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See credit lines of images or other third-party material in this article for license information.



Keywords

Open Targets, therapeutic hypothesis, genetic evidence

Abstract

Open Targets, a consortium among academic and industry partners, focuses on using human genetics and genomics to provide insights to key questions that build therapeutic hypotheses. Large-scale experiments generate foundational data, and open-source informatic platforms systematically integrate evidence for target–disease relationships and provide dynamic tooling for target prioritization. A locus-to-gene machine learning model uses evidence from genome-wide association studies (GWAS Catalog, UK BioBank, and FinnGen), functional genomic studies, epigenetic studies, and variant effect prediction to predict potential drug targets for complex diseases. These predictions are combined with genetic evidence from gene burden analyses, rare disease genetics, somatic mutations, perturbation assays, pathway analyses,

scientific literature, differential expression, and mouse models to systematically build target–disease associations (<https://platform.opentargets.org>). Scored target attributes such as clinical precedence, tractability, and safety guide target prioritization. Here we provide our perspective on the value and impact of human genetics and genomics for generating therapeutic hypotheses.

THE OPEN TARGETS CONSORTIUM

The overall success rates in drug discovery are low; 90% of drugs that enter phase 1 clinical trials do not reach the market due to insufficient efficacy or safety issues that surface in later, expensive clinical trials. This failure rate drives the overall cost of drug development, which is currently estimated at between \$1 billion and \$2 billion for each approved medicine (1–3).

Drug discovery begins with the selection of a target that is proposed to be involved in the disease process and the construction of a therapeutic hypothesis around that target describing an intervention that will benefit the patient. Target identification and prioritization are a complex problem encompassing how we understand disease processes, how we predict the impact of perturbing our complex biology, and how we select from a range of potential options the approach with the highest chance of success in new drug discovery. The rate of late-stage clinical failures shows that there is a huge opportunity to improve our capabilities to select the targets that can lead to safe and effective medicines.

In 2014 discussions between the Wellcome Sanger Institute, European Molecular Biology Laboratory’s European Bioinformatics Institute (EMBL-EBI), and GSK led to the founding of the Centre for Therapeutic Target Validation. This innovative, large-scale, multiyear, public–private partnership was created to use human genetic and genomic data for systematic drug target identification and prioritization. Rebranded as Open Targets in 2016, the partnership has grown and now brings together the expertise of EMBL-EBI and the Wellcome Sanger Institute with industry partners GSK, Pfizer, Sanofi, Bristol Myers Squibb, and Genentech to enhance the translational impact of academic science on human health and the capabilities of industry to envision new, safe, and effective medicines (**Figure 1a**).

Working in partnership across academia and industry is one of Open Targets’ core philosophies, bringing together the best of academic and industrial approaches. Projects are designed and executed and results analyzed by teams of academic and industry scientists who stay connected through the life cycle of each project to deliver impactful insights that improve the target identification and prioritization landscapes (**Figure 1b**). Open Targets research is made available to the community through publication, and our approach to target identification and prioritization is exemplified by our flagship open-source informatics platforms, Open Targets Platform (<https://platform.opentargets.org>) and Open Targets Genetics (<https://genetics.opentargets.org>) (4, 5). We and others have shown the enrichment of targets supported by genetic evidence in successful drug discovery programs (6–8). Genetic evidence for a target–disease association is fundamental to building a strong therapeutic hypothesis. In this review we outline our perspective on using human genetics and genomics for drug target identification.

THE OPEN TARGETS RESEARCH PROGRAM

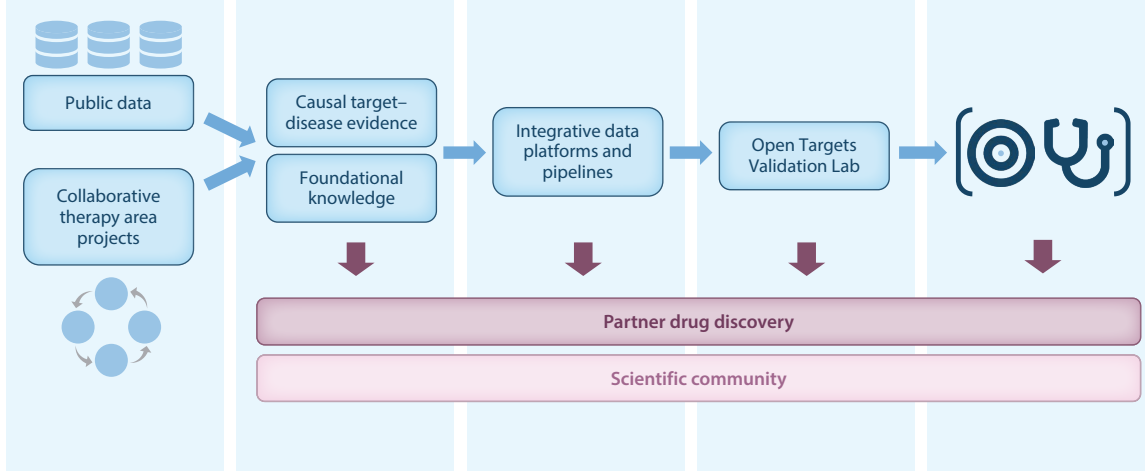
To achieve the vision of Open Targets, the consortium research program focuses on systematically generating and understanding relevant data that enables us to identify and prioritize targets for drug discovery. This requires the generation of causal target–disease evidence alongside

foundational disease and biological process understanding so that we can decipher the role of potential drug targets through the life cycle of disease processes. To build therapeutic hypotheses from target–disease relationships, we need to be able to understand them in the context of the biology of the pathways, cells, and tissues involved in disease and then propose specific targeted interventions that will be beneficial for the disease state in question (**Figure 2**). The Open Targets approach is aimed at understanding the data that connect targets to disease, through contextual

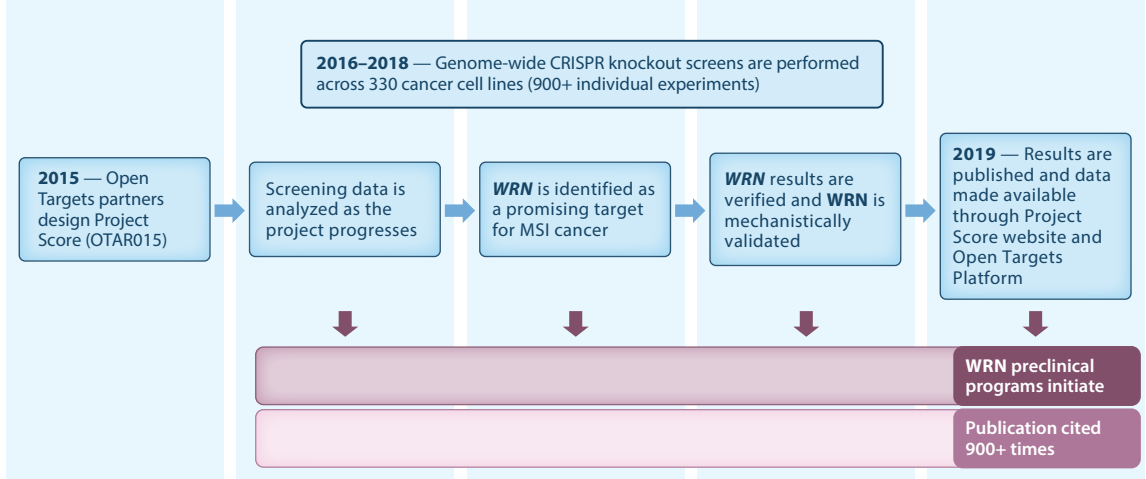
a All Open Targets partners collectively actively design, prioritize, and participate in the research program



b General Open Targets project framework to facilitate systematic target identification and prioritization for safer and more effective treatment of disease



c Example: Timeline for Werner Syndrome ATP-dependent Helicase (*WRN*), identified as a target for tumors with microsatellite instability



(Caption appears on following page)

Figure 1 (Figure appears on preceding page)

An overview of how the Open Targets consortium works. (a) The Open Targets academic partners [European Molecular Biology Laboratory's European Bioinformatics Institute (EMBL-EBI) and Wellcome Sanger Institute] and industry partners [GSK, Bristol Myers Squibb, Sanofi, Pfizer, and Genentech (a member of the Roche group)] work together to collectively design and prioritize the research program and actively participate in projects and platforms that are developed. The collective vision of the consortium is to systematically identify and prioritize targets for the development of safer and more effective treatment of disease. (b) An illustration of the Open Targets project framework. We bring together publicly available data with data generated through the collaborative research program's projects to provide causal evidence for target–disease associations as well as foundational knowledge related to genes and diseases. This knowledge is integrated into our open-source informatics platforms. Potential targets are fed into our validation lab for further mechanistic insights. This workflow at each step of the way helps feed into our industry partners' drug discovery pipelines and, once publicly available, into the wider scientific and drug discovery community, with the ultimate aim of helping to develop safer and more effective medicines. (c) An example of an Open Targets research project showing the emergence and journey of a potential target lead through the workflow. Project Score was designed as a collaborative research project to comprehensively knock out genes in 330 different cancer cell lines. Analysis and target prioritization revealed *WRN* as a promising novel candidate in microsatellite instable (MSI) colorectal cancer. This was further mechanistically validated. The results from the study were published (20), made open source through the Project Score website (<https://score.depmap.sanger.ac.uk>) (23), and integrated into the Open Targets Platform to provide target–disease associations (<https://platform.opentargets.org/>), feeding into preclinical drug discovery programs in industry and providing knowledge to the broader scientific community.

knowledge that helps build therapeutic hypotheses and ultimately catalyzes systematic, evidence-based decision-making that will lead to new, safe, and effective therapies.

The Open Targets research program comprises two intertwined pillars, spanning experimental and informatics activities with a focus on systematically generating and interpreting data that enable us to enhance the target nomination for drug discovery. The initial step in this process is building a confident causal gene–disease linkage (**Figure 2b**). To achieve this, we systematically ingest, harmonize, and integrate foundational publicly available datasets together with the data generated through internal projects (9–12) (**Figure 1b**). The target–disease link is then contextualized to provide information on cell type and tissue expression as well as biological processes (13). The complementarity of the information captured through these processes enables us to take a systematic approach from genetic association toward developing a therapeutic hypothesis (**Figure 2c**). Finally, the emerging hypotheses are evaluated through the lens of drug discovery, defining what can be done with the available drug modalities, what option will be both efficacious and safe, and what option is best positioned against current standards of care and other treatments in development (**Figure 2c**).

The field of drug discovery has undergone a profound transformation with the advent of genomics. Traditional drug discovery approaches often relied on phenotypic screens, but the integration of genomic data into this process has shifted the paradigm toward a more informed and targeted process. The generation of foundational genetic and functional genomic data coupled with genetic engineering screens and subsequent analyses promises accelerated insights into disease mechanisms and identification of novel therapeutic targets. The process can involve large-scale sequencing of genomes, transcriptomes, and epigenomes from individuals with the target disease and/or healthy controls or patient-derived cells that are systematically studied in vitro in the context of specific disease-relevant cellular processes.

Genome-wide association studies (GWAS) have been instrumental in identifying genetic variants associated with various diseases. Systematic ingestion and harmonization of GWAS results form the core of our ability to formulate causal gene–trait links (5, 9, 14, 15). For diseases of strategic interest to our consortium, such as inflammatory bowel disease (IBD), we further undertake dedicated efforts to systematically fine-map GWAS variants to increase our confidence in defining causal genes (16). Furthermore, we develop novel approaches that enable us to experimentally fine-map variants, such as an arrayed CRISPR screening method referred to as genome

a Therapeutic hypothesis concepts

Causal and supporting evidence

Considerations for target–disease association and understanding direction of effect and mechanism to enable therapeutic hypothesis construction

Precision and context

Considerations for the precision and personalization needed for effective therapy for defined biological hypothesis, disease stage, and defined patient group including tissue and cellular context, cell and cell state, disease subtype and state, and patient stratification

Hypothesis progressability

Considerations for the potential to progress a drug discovery program, including clinical precedence, modality and tractability, doability, and safety around the therapeutic hypothesis

b Target identification

What evidence associates this target to this disease?

What targets have genetic evidence?
What evidence is there from other sources?

- What are the likely causal genes underlying associations for my disease of interest from a large population biobank genome-wide association study?
- Is there evidence from rare variation (gene burden analysis)?
- Is there genetic evidence from clinical sources?
- Is there evidence from perturbation screens to suggest a role of this gene in this disease?
- What targets are extracted from the scientific literature and patent information with an association to my disease/trait of interest?
- What targets are differentially expressed between normal and disease tissue?
- Are there proteins in the same pathway or interaction network as a known target?
- By broadening the disease ontology landscape, can I identify new targets?
- Is there any evidence from animal models?
- If my disease of interest is oncology related, is there somatic mutation evidence or evidence this target is a cancer hallmark/driver?

c Target prioritization

Tractability

- Should I inhibit or activate the target?
- What type of modulator would be best?
- Are there predicted or known binding pockets in this target?
- Is the target accessible?
- Which cell types are relevant for this disease and is my target specific for this cell state?
- Where should I modulate the target to get the most benefit for this disease?

Doability

- Is there a probe that binds my target?
- Is there a suitable assay or cell model for measuring target effect?
- Is there homology for my gene and similar expression in preclinical model species?
- Is there a suitable animal model for my target and disease of interest?
- Is there a biomarker I can use to measure target modulation or effect on the disease?

Safety

- What are the possible risks in modulating this target?
- Have genetic variants in this target been associated with adverse drug events (pharmacogenetics)?
- Are there alternative targets in the same biological pathway that would be safer?
- Are there fewer loss of function variants than would be expected (genetic constraint)?
- What is the effect of modulating this target on the system as a whole?
- Have adverse drug events been reported for drugs that modulate this target (pharmacovigilance)?
- Which patients are likely to respond/have side effects?
- Is this gene specifically expressed in the relevant tissue or cell type of interest?
- Have unwanted/severe phenotypes been reported in a mouse model associated with this target?
- Is this gene essential?

Clinical precedence

- Are there already drugs for my target/for my disease of interest?
- Are there compounds in clinical trials that modulate this target?
- What is the unmet clinical need for this disease?
- Would this be a novel target/novel mechanism?

Additional considerations

- Why do nonresponders to current drug treatment fail to respond?
- Which targets are relevant for early-onset disease versus late onset?
- Are there suitable targets on a different pathway that could be used in combination therapy to avoid resistance?

Therapeutic hypothesis generation

(Caption appears on following page)

Figure 2 (Figure appears on preceding page)

The Open Target therapeutic hypothesis-building framework. (a) The Open Targets framework for building therapeutic hypotheses is composed of three overarching concepts that all contribute to a fuller picture of evidence and context to help in the decision-making process for taking a potential target forward for drug discovery. In blue are considerations for target–disease association and understanding the causality of disease, mechanism, and direction of effect. In yellow are considerations for effective therapy that include cell and tissue context, disease subtype and state, and patient stratification factors. In red are considerations for the progressability of a target, which include how tractable a target is, suitable modalities, how safe it may be to modulate it, whether there are tools available for the next stages of validation or preclinical studies (doability), and the clinical precedence for the therapeutic hypothesis that is being developed, for example, whether known drugs for the target are in current clinical trials or have been approved. Throughout this framework, answers to questions provide evidence to help build up a therapeutic hypothesis, first by identifying targets associated with a given disease of interest (b), then through questions that help further prioritize these targets (c). Through the Open Targets research program and informatics resources we generate evidence and evaluate the evidence to help answer the questions that are important for target identification and prioritization for drug discovery. In this figure we provide some examples of the types of questions we try to address in order to help build therapeutic hypotheses. This framework enables the construction of therapeutic hypotheses such as “inhibition of target X will be beneficial in the treatment of disease Y for patients of disease subtype Z, in cell type A, in tissue B, with drug modality of C. Effective modulation of Target X can be monitored by biomarker D.” An example from results of Project Score is: “inhibition of WRN may be beneficial in the treatment of colorectal cancer, for patients with a microsatellite instable tumor subtype” (20, 23). In the main text we provide more detail on how genetics and genomics help provide answers to these questions, and how we have developed the Open Targets Platform to aid therapeutic hypothesis construction. (b) Target identification is the first step in building a therapeutic hypothesis and involves evaluating the evidence for target–disease associations. Example questions that may be asked are outlined here; answers to these questions can be explored in the Open Targets Platform utilizing human genetics and genomics along with other data types (<https://platform.opentargets.org/>). (c) Once a list of targets for a given disease has been established and prioritized based on weighing up of different evidence types, questions related to precision for effective therapy and progressability of targets become important for prioritizing targets and developing a full therapeutic hypothesis.

engineering-based interrogation of enhancers (GenIE), which assesses the effects of defined alleles on transcription or splicing when introduced in their endogenous genomic locations (17). We applied GenIE to screen the 99% credible set of Alzheimer’s disease (AD) GWAS variants clustering at a locus to identify a subset of likely causal variants in human induced pluripotent stem cells (iPSCs) (12).

Disease-associated variants highlight gene and biological pathways that play a critical role in disease development, and leveraging genetic signals can help pinpoint the cell type and cell states most relevant for a disease and identify causal genes. For example, variants associated with immune-mediated diseases have been implicated in regulating gene expression in T cells (18). We have systematically profiled chromatin activity in naive and memory CD4⁺ T cells and monocyte-derived macrophages exposed to different cytokine conditions mimicking proinflammatory states. Using this dataset of 55 cell states, we developed a novel statistical framework for testing single nucleotide polymorphism enrichment within chromatin regulatory elements. We identified that immune-disease variants were especially enriched in the early activation of memory T cells (19). Homing down on this disease causal process, we then generated a single-cell RNA sequencing (scRNA-seq) map of expression quantitative trait loci (eQTLs) during a time course of T cell activation. This approach enabled us to identify 127 causal genes whose expression is regulated by immune-disease-associated variants (11). The findings from this study have implications for drug target discovery. Proof-of-concept examples include immune-disease variants regulating the expression of genes such as *CTLA4* and *TYK2*, which are targets of existing therapeutic interventions.

The emergence of CRISPR-based genetic screening technologies has started a new era in mapping regulators of cell functions and, as such, offers an accelerator for drug discovery. By systematically interrogating the function of individual genes, CRISPR screens enable researchers to uncover crucial genetic dependencies and vulnerabilities within various diseases, providing valuable insights that were previously inaccessible. CRISPR screens have empowered researchers to

comprehensively catalog genetic dependencies across diverse cellular contexts. These screens, spanning CRISPR knockout and activation libraries, have elucidated essential genes, signaling pathways, and molecular networks critical for cell function regulation. This understanding forms an important milestone for targeted drug development. Perturbation screens generated through our research program complement other publicly available large-scale screens in the Open Targets Platform such as CRISPRbrain (<https://crisprbrain.org>) to provide target–disease evidence and inform on target modulation and safety.

Project Score is an exemplar of how Open Targets research projects are created, managed, and ultimately shared with the scientific community (**Figure 1c**) (20). In 2015 Open Targets held a call for projects in which a team of academic researchers and industry scientists designed an ambitious program of genome-wide CRISPR knockout screening in cancer cell lines to systematically identify synthetically lethal targets across different cancer types. Capability was built both experimentally and informatically to perform and analyze more than 900 individual screening experiments conducted across 330 cancer lines between 2015 to 2018. Academic and industry consortium partners worked together to analyze and interpret the results and build an overall landscape of targets that incorporated the synthetic lethality results with other available data as a prioritization score assessed against the predicted tractability of the targets (20).

As the project generated data over time and the analysis progressed, the screens identified the critical role of *WRN*, a gene encoding a RecQ family DNA helicase, as a potential therapeutic target in microsatellite instable (MSI) colorectal cancer. Open Targets partners worked together to define a validation plan for *WRN* to verify these finding and undertake mechanistic validation of the mode of action (**Figure 1c**). These studies discovered that *WRN* depletion induces DNA double-strand breaks, cell cycle arrest, and apoptosis, specifically in MSI+ cells. *WRN* was independently identified as a promising target simultaneously by another group (21), and the findings are now being further followed up in preclinical investigations outside of the Open Targets consortium to develop potential therapies targeting *WRN*'s helicase activity, providing promising avenues for novel cancer therapies. The project was published in 2019 (20), and the data were shared interactively with the scientific community as the Project Score database providing integrated CRISPR screen data, genetic biomarkers, and target tractability assessments (<https://score.depmap.sanger.ac.uk>) (22, 23). These data are incorporated into the Open Targets Platform, and the Project Score database has since progressed independently within the DepMap Consortium to integrate similar screening results performed by other groups (<https://depmap.org/portal>). This project demonstrates how Open Targets works together with its academic and industry partners to design, execute, and ultimately make available to the community large-scale, systematic studies to inform target identification and prioritization (**Figure 1c**).

Open Targets' approach encompasses systematic large-scale experiments and focused studies generating in-depth knowledge in our strategically important therapy areas (oncology, immunology and inflammation, and neurodegeneration). Incorporating genomic data into drug discovery is revolutionizing the field by providing a solid foundation of genetic knowledge and enabling functional insights into disease mechanisms. The combination of foundational genetic data and functional analyses empowers researchers to identify and prioritize novel therapeutic targets with a higher likelihood of success. The Open Targets approach is aimed at helping interpretation of the data that connects targets to diseases through contextual knowledge, moving the drug discovery dial from genetic discovery toward building therapeutic hypotheses, and ultimately catalyzing systematic, evidence-based decision-making that will lead to new, safe, and more effective therapies.

VARIANT TO GENE TO FUNCTION FOR TARGET IDENTIFICATION

Drugs that have been developed to modulate a target that have supporting GWAS evidence are twice as likely to succeed and be approved, and those with underlying Mendelian genetic evidence are up to eight times more likely to succeed (6, 24). We have shown that two-thirds of the drugs approved by the US Food and Drug Administration (FDA) in 2021 have underlying genetic evidence for the target–disease association (7), and that this trend extends across the past decade (8). Genetic evidence for the association between a target and a disease is therefore of key importance for identifying and prioritizing targets for novel drug development; the challenge arises with connecting evidence from GWAS to a target. The majority of associations from GWAS are between genetic variants that are not in a coding region; the reported lead variant may be in linkage disequilibrium with the true causative genetic variant, or the variant may be influencing gene regulatory mechanisms. Furthermore, the genetic association comes with no effector cell type or tissue context. There is therefore a need to connect GWAS to the likely disease-causing gene; to do this in a systematic way for all GWAS loci is complex.

Our approach at Open Targets has been to develop a machine learning (ML) model to provide a predictive locus-to-gene (L2G) score for the most likely causative genes underlying each GWAS association (14). We provide a summary of the complex analysis pipeline below; however, more detailed documentation and an illustration are available at <https://genetics-docs.opentargets.org/our-approach/prioritising-causal-genes-at-gwas-loci-l2g> and are described in our methodology publication (14). In total, 57 locus-level predictive features are defined for ML, providing a scored rank of likely causative genes (targets) underlying each GWAS locus. The main categories of evidence sources for the predictive features are the following:

1. Gene distance to credible set variants (weighed by their fine-mapping probabilities)
2. Molecular quantitative trait locus (QTL) colocalization
3. Chromatin interaction
4. Predicted variant pathogenicity

A fundamental foundation for the L2G pipeline is the integration and harmonization of large-scale genetic and functional genomic datasets and subsequent statistical analyses, which provide the evidence input for the predictive features of the ML model. GWAS data from EMBL-EBI's GWAS Catalog (9), UK Biobank (25) (<https://www.nealelab.is/uk-biobank>), and FinnGen (<https://www.finnngen.fi/en>) are harmonized and processed [including mapping of all traits to EMBL-EBI's Experimental Factor Ontology (EFO) (26)]. Systematic fine-mapping is performed from available summary statistics to generate credible sets of causal variants across all the lead variant-associated loci (FinnGen fine-mapping is provided separately). Trait–trait colocalization analysis is carried out for GWAS datasets with summary statistics, which are provided by the GWAS Catalog (9, 27). Tissue and cell type–specific molecular (QTL) datasets are integrated and used to carry out trait–molecular trait colocalization. These include eQTLs (10, 28, 29), protein QTLs (30–36), and splice QTLs (10). In total, over 3.7 million colocalization tests are performed and provided. The trait–molecular trait colocalizations are used as a predictive feature for the ML model. In silico functional predictions from Ensembl Variant Effect Predictor (37) and PolyPhen, epigenomic information including promoter capture Hi-C and DNase hypersensitivity sites (38–40), and gene distance to credible set variants weighed by their fine-mapping probabilities also provide data inputs for the predictive features.

To create a gold standard set of genes from solved GWAS loci that can be used as a training set for the ML model, our approach was to manually curate a set of gene–disease or gene–trait associations of high confidence level. High confidence for the initial gold standard GWAS gene assignment list of 445 genes for the first release was classified based upon known drug

target–disease pairs (41, 42), experimental alterations from literature reports (for example, nucleotide editing), and observational functional data (for example, colocalizing molecular QTLs, colocalizing epigenetic marks, reporter assays) (43); it also included expert domain knowledge of strong orthogonal evidence or biological plausibility, curation from the T2D Knowledge Portal (<https://t2d.hugeamp.org>), and Open Targets curated associations. We continue to update this list with new knowledge and include known target–disease associations for approved drugs and associations sourced from the scientific community. The gold standard list is made publicly available for the community to utilize and feeds back into community efforts to establish known effector genes in disease (<https://github.com/opentargets/genetics-gold-standards>).

The L2G model produces a score ranging from 0 to 1; a gene scored with 0.5 can be interpreted as having a 50% chance of being the causal gene at the locus (under the assumption that the characteristics of the observed locus are similar to those in the positive gold standard set). The integrated data, analyses, and L2G score are made available in Open Targets Genetics, which serves as a comprehensive publicly available source for post-GWAS analyses (5).

Data availability for post-GWAS analysis is a challenge, and the first issue we wanted to overcome was the availability of summary statistics from GWAS, required in order to carry out fine-mapping and colocalization. Summary statistics provide aggregate p values and association data for every variant analyzed in a study. With EMBL-EBI being one of the Open Targets partners, we were able to collaborate closely with the GWAS Catalog team to establish a pipeline for the submission of full summary statistics from GWAS, which are available via an application programming interface (API) or download from the GWAS Catalog (27). This involved harnessing the scientific as well as the publishing community to advocate for the submission of full summary statistics, not only for use by Open Targets, but for the community as a whole. This has had a large impact; since the release of the author submission system in 2020, studies with summary statistics have grown from approximately 1,000 studies to over 55,000 studies in the GWAS Catalog (27). Similar challenges exist for using population biobank resources, where genotype-level data are not readily made available for fine-mapping, for example. Our approach has been to collaborate closely with FinnGen, which provides its own fine-mapping for us to use internally (<https://www.finnngen.fi/en>).

There is a gap in available funding opportunities for new databases and bioinformatics tooling, and one of the advantages of the Open Targets consortium is the ability to identify the need and collect the requirements for a new resource from our own target identification and prioritization perspective, as well as within the wider scientific community. We are able to support the generation of new open-source resources that help bridge an existing gap in the scientific community. An example is the EMBL-EBI eQTL Catalogue, which was developed to ingest and uniformly compute gene expression and splicing QTLs from different published studies (10). This enables us to efficiently utilize the harmonized data for colocalization with GWAS loci as part of Open Targets Genetics pipelines, which ultimately allows us to prioritize targets from large-scale GWAS data. This useful resource is also available to our partners as well as to the scientific community as a whole. We continue to support and expand the scope of the eQTL Catalogue as the functional genomic research field develops in complexity, working with the single-cell eQTLGen Consortium toward harmonizing the integration and representation of single-cell eQTL data (44).

SYSTEMATIC INTEGRATION OF TARGET-DISEASE EVIDENCE FOR TARGET PRIORITIZATION

There is an ever expanding and wide range of different human genetic and genomic data that can be used to inform on the role of a potential drug target in a given disease. Integration,

harmonization, and interpretation of this wealth of big data for target discovery and prioritization are complex. Our approach at Open Targets has been to create a comprehensive open-source resource that would bring together relevant publicly available data in an intuitive way to help identify and prioritize targets for drug discovery (4). The Open Targets Platform systematically prioritizes targets [represented as genes (45)] across all diseases [mapped to EFO (26)] and weighs evidence within and across different data types using a harmonized sum (4). Weights of the contributing evidence sources can be adjusted according to user requirements, which in turn dynamically adjust the prioritization of targets. Due to the importance of genetic evidence for supporting successful drugs, resources that provide germline genetic associations (including the L2G score for target–disease associations from Open Targets Genetics) form a data type prioritization category and are a key contributor to the overall prioritization score. Associations from somatic mutations also form a separate data type category. Other data source type categories include known drugs, pathways/systems biology, text mining, RNA expression data, and animal models. Each data type can be selected as required, allowing users, for example, to require contributing genetic evidence for target–disease associations. This enables users to answer the types of questions outlined in **Figure 2b** for target identification and the initial prioritization of a list of targets for their disease of interest, presenting the underlying evidence from each data type.

There is a growing wealth of curated data for associations between genes and rare diseases. Our approach has been to harness this knowledge in several ways to provide insights into complex multifactorial diseases. In the Open Targets Platform we integrate gene–disease validity assessments from multiple resources (ClinGen, PanelApp, Gene2Phenotype, Orphanet) (46–49), using Gene Curation Coalition terminology where possible (50), and use the assessment of evidence to weigh the contributing association score. The increasing number of clinical whole exome and genome sequencing studies has enabled the development of gene burden analysis methods to assess the contribution of rare variation across a given gene to complex traits. We have integrated the results of gene burden analysis from multiple studies in the Open Targets Platform, continually updated with major published studies that are released (51–61). We have investigated the integration of gene burden analysis results both into the gold standard list and as a feature in the L2G model for Open Targets Genetics.

The evidence from rare disease curated sources is combined with the complex disease genetic evidence from the Open Targets Genetics L2G score and gene burden analyses to provide genetic association scores for target–disease associations in the Open Targets Platform. This allows questions as exemplified in **Figure 2b** to be explored through contributing evidence types. All diseases/traits are mapped to EFO, enabling the exploration of evidence for similar traits or diseases. Evidence for target–disease associations from somatic mutation resources (the Catalog of Somatic Mutations in Cancer’s Cancer Gene Census, IntOGen, ClinVar) is categorized separately (62–65). As a result, the role and pathologic mechanism of a particular target in particular diseases become apparent; for example, *KRAS* has strong germline genetic evidence (rare disease resources and L2G) and no somatic mutation evidence for the association with RASopathies, whereas the opposite is true for the association with non–small cell lung carcinoma, lung adenocarcinoma, and colorectal adenocarcinoma (for which inhibitors targeted at a specific somatic residue in *KRAS* have been developed). In the future, we hope to extract more information on the direction of effect involved from these genetic resources, in order to inform on the mechanism of action suitable for a particular therapeutic and to inform on potential drug repurposing opportunities for diseases of unmet clinical need. This can also be valuable knowledge in the prediction of possible side effects. For example, knowing that loss-of-function variation in a particular gene results in a severe phenotype may provide caution as to the use of an inhibitor.

ENABLING THERAPEUTIC HYPOTHESIS BUILDING

The foundation for enabling therapeutic hypothesis building in the Open Targets Platform is the association between a gene (target) and a given disease and the extent of the evidence for this (**Figure 2b**). We are then building upon this to provide a comprehensive picture to enable the development of therapeutic hypotheses based upon answers to questions that lead to further knowledge around disease mechanism, context, and progressability (**Figure 2a,c**). This allows further prioritization of targets to indicate whether it is sensible to move this target forward into drug discovery pipelines, as well as what therapeutic type it may be most suitable for.

The Link Between Disease Mechanism and Therapeutic Modality

Genetic data can inform on the likely direction of effect of the disease association and provide insight into the mechanism. This in turn can inform on the likely desired direction of effect of a given therapeutic agent, such as the use of an inhibitor against a target in which gain-of-function variants cause disease (66). This goes hand in hand with assessing safety because it also provides information about the possible undesired effects of inhibiting the target where there is evidence from human cases of loss of function resulting in a severe rare disease phenotype, in vitro perturbation assays, or in vivo animal knockout models. We plan to build upon this through our research program to provide dose response curves for targets across a range of phenotypes. The direction of effect is phenotype-specific, so weighing these attributes with regard to the therapy area becomes important. For oncology, for example, the ultimate goal is to kill the cancer cells, and because of the usually urgent and short-term treatment protocol for patients, adverse effects are often tolerated more than with chronic inflammatory diseases, which require long-term treatments along with a need to dampen down damage to tissue and help restore cellular function.

Target Tractability

In the Open Targets Platform, we provide tractability assessments of targets for the suitability of modulating a target with a small molecule, antibody, or proteolysis-targeting chimera (PROTAC) using different key attributes that would be required for each type of modality (67). For example, if there is a high-quality binding pocket a small molecule may be suitable, a location on the cell surface is important for the use of an antibody, and ubiquitination is important if considering a PROTAC. Clinical precedence feeds into these assessments, such as whether a therapeutic of that modality has already been developed and what clinical trial phase this has reached. As the expanding range of new modalities opens up new therapeutic possibilities, in accordance with the theory that no target is undruggable, we will continue to identify key features that help evaluate the suitability of a target for a particular modality to expand our tractability assessments in the Open Targets Platform. We also will enhance the use of data such as binding pockets utilizing structural protein predictions from AlphaFold (68, 69). New regulatory pathways for drug approval make drugs designed for a particular patient subgroup more attractive and feasible, and genetics can be utilized for patient stratification or for target strategy. A key recent example in oncology is Sotorasib, which targets *KRAS*^{G12C} to treat non-small cell lung cancer patients with this mutation (7). In rare disease there is an expanding portfolio of antisense oligonucleotides that target the underlying causal genetic variant in the pre-mRNA, for example, in the treatment of subgroups of Duchenne muscular dystrophy patients with variants amenable to exon 45 (casimersen), 51 (eteplirsen), or 53 (golodirsen and viltolarsen) skipping (8, 70).

Doability

At Open Targets, doability involves features that enable a drug discovery program to be readily pursued through validation, for example, the existence of a publicly available chemical probe, a mouse ortholog, or a knockout model that can provide the basis of further study of the function of the target. Expansion of the pharmacological tool kit through efforts such as Target2035, which aims to provide a chemical and/or biological probe for the entire human genome by 2035, will open up the ability to explore a wider range of targets (71). A known biomarker for assessing target engagement and effect on biological mechanism is important for being able to take a target forward, and we plan to provide this information where available.

Safety

The potential safety of modulating a target is a key factor for prioritizing some targets over others, and if there is evidence that could cause major concern, a target will not be taken forward. There are many different sources of evidence for whether it would be safe to modulate a target (and to what extent) that can be considered prior to carrying out extensive and expensive preclinical studies. At Open Targets, the approach has been to establish an informatics research project around safety and toxicity evaluation for targets, for which different workstreams feed useful annotations into the Open Targets Platform (**Figure 1b**) and help answer questions such as those outlined in **Figure 2c**. This includes curated assessments from key published reviews, publicly available safety-related databases, and adverse outcome pathways (72–77). From these resources, the safety event associated with modulation of the target, the reported affected biosystem, the direction, and the dosing effect on the target (for example, activation or inhibition, acute or general) are provided. Genetics is also informative for safety assessment (78), and we have several approaches to provide genetic-related information for target safety assessment in the Open Targets Platform. Possible safety effects of perturbing or inhibiting a target can also be inferred from the gene essentiality and genetic constraint. On target pages in the Open Target Platform, we provide gene essentiality assessments across 28 tissues based upon cell fitness after CRISPR knockout of the gene in different cell lines from the Cancer DepMap (22). A gene is considered a core essential gene if the majority of cell lines do not survive after target knockdown; this provides an indication of how target inhibition is likely tolerated. Whether a target is a known cancer driver or a tumor suppressor gene can also provide important information to assess whether modulation would be appropriate. These are potentially good targets in cancer; however, inhibition of core essential genes and tumor suppressors has likely consequences for adverse effects. The genetic constraint assessment from the Genome Aggregation Database (<https://gnomad.broadinstitute.org>) (79, 80) provides a natural population-level perspective of how safe it may be to perturb a target; targets that have very high genetic constraint (with much less loss-of-function variation than would be expected) may be more risky to perturb than those with low constraint, which have shown tolerance to loss-of-function variation over evolution. Comparative genomics can also provide safety information; knowing whether there are human paralogs with strong homology to a given target can provide an indication of the risk of interference with other targets. The specificity and distribution of the target are important considerations for the likelihood of on-target effects beyond the disease-causing tissue and the possible severity of these (such as effects on the central nervous system or cardiovascular system versus liver toxicity). The likelihood of such effects is indicated by assessments from RNA baseline expression across tissues, which are provided by Expression Atlas and GTEx (28, 81).

At a whole organism level, we provide an assessment of the severity of knocking out the gene based upon classification of phenotypes from mouse models available in the Mouse Genome

Informatics database, which gives an indication of the severity of possible side effects when modulating the target (82). On a drug level, we also provide pharmacovigilance assessments analyzed using data from the FDA Adverse Event Reporting System (<https://open.fda.gov/data/faers>) and black box warnings from drug labels (41, 42, 83). Though these may be off-target events, they can provide useful real-world clinical and systemic safety information regarding the effect of modulating a given target in humans.

Our Approach to Artificial Intelligence and Machine Learning

Open Targets uses artificial intelligence (AI) to address specific scientific challenges such as identifying the most likely target genes underlying GWAS associations; for this, we developed the L2G ML pipeline to provide predictive scoring of candidate genes (14). We apply AI to knowledge extraction in several ways; for instance, named entity recognition and deep learning are used to extract publications and patents related to genes, diseases, drug entities, and relationships that can be explored through the Open Targets Platform in collaboration with Europe PMC (<https://europepmc.org/>) (84). We also have developed natural language processing to categorize why clinical trials have failed (informing on target–disease–drug associations that have resulted in lack of efficacy or safety issues) (85), and to mine drug label information for clinical indications in collaboration with EMBL-EBI's ChEMBL database (41). This provides information on clinical precedence and safety (**Figure 2c**). We explore new technology and applications across our research program and platforms, for example, by using OpenAI's large language model to provide a contextual summary of the relationship between a target and a disease in a given publication.

Open Targets is actively developing its dataset into a unique repository of high-quality data with well-established data provenance, often linking it to other datasets using unified identifiers to facilitate downstream analyses. Our datasets are utilized externally as a gold standard for a wide array of AI applications, such as ML to identify novel target–disease associations (86), building knowledge graphs for a range of different biological insights and to assist with drug development (87–90), and benchmarking new computational methods for drug target prioritization (91, 92). There are examples of pharmaceutical companies beyond Open Targets partners who have utilized data sourced from Open Targets to build their own target prioritization and drug repurposing pipelines (93). Additionally, the data from Open Targets play a central role in establishing the foundations of research data and are soon set to become the backbone of foundation model development. Furthermore, the data and pipelines empower the creation of various data lakes and the development of end-to-end pipelines for ingesting/generating, integrating, and analyzing multiomics data.

The Benefits of Being Embedded Within EMBL-EBI and the Wellcome Sanger Institute

Because data are integrated into Open Targets resources using a consistent mapping of traits/diseases to EFO, we can integrate different data sources and collate evidence for the same disease that otherwise would not have been connected. This also enables the exploration of associations with targets up or down the ontology hierarchy [for example, connecting all targets related to small bowel Crohn's disease (CD) to immune system disease]. We work closely with other EMBL-EBI resources to develop the requirements for this systematic integration. This includes the expansion of terms in the EFO (26) and the development of specific data feeds or data formats from resources such as EMBL-EBI's ChEMBL database (for drug label indications and clinical trial information) (41, 42), the European Variation Archive (for ClinVar annotations) (94), and the Expression Atlas

(95). We are able to request focused curation for disease areas of particular interest to our partners and harness our scientific community to help review and provide feedback to the resources. An example of this is Reactome's in-depth curation of the autophagy pathway, chosen due to its role in Open Targets' key therapeutic areas and the need for a comprehensive view of the mechanistic role of proteins involved in the process and interaction with other pathways (96). Through the Open Targets consortium community, we were able to harness expert review of the pathway and provide useful connections to Reactome through the Open Targets Platform. Being embedded within the Wellcome Sanger Institute, we have access to high-throughput experimental platforms as well as vital patient samples through close collaborations with clinical networks within the National Health Service England. This also provides the infrastructure to store, process, and analyze samples safely.

PARTNER AND COMMUNITY IMPACT

The Open Targets consortium is an innovative partnership between academia and industry featuring continuous collaboration and knowledge exchange between all partners. Academic partners contribute deep expertise and innovation in genetic and genomic research, while industry partners provide an understanding of real-world applications and challenges faced in drug discovery and development. A robust and responsive feedback loop ensures strategic alignment and maximizes impact for industry partners and the broader scientific community.

Open Targets generates impact through several avenues. First, it supports the collation of publicly available data into large databases of harmonized datasets (e.g., through supporting the development of the summary statistic submission portal of the GWAS Catalog and establishing the eQTL Catalogue) (9, 10, 27). This addresses a major gap in the scientific community in which the abundance of data available in the public space is often dwarfed by the effort needed to process and store the data in a format that enables broader use.

Second, Open Targets academics push the boundaries of science by developing novel experimental and in silico approaches. This is achieved through bespoke Open Targets projects that are formulated jointly by all partners (**Figure 1**). These projects leverage the unique capabilities of Open Targets, particularly the cutting-edge genomic and bioinformatic capabilities of the Wellcome Sanger Institute and EMBL-EBI, and influence target discovery and validation work in industry partners. Prepublication access to data and results from several Open Targets projects has been crucial for specific drug discovery efforts. One of these projects generated a large-scale scRNA-seq dataset from the terminal ileum of CD patients. Uniquely, this study also has germline genotype data on these CD patients (15). In addition to the in-depth analyses done by the project team, this allowed partners to do their own bespoke analyses, which included running additional methods to integrate scRNA-seq with genetic data to further elucidate the mechanisms contributing to IBD risk at the cellular level.

Finally, all insights are integrated into the comprehensive, open-access Open Targets Platform, which is extensively used in industry to facilitate more efficient decision-making, from the identification of drug targets to the design of clinical studies (**Figure 2**). Efficient access to multiple harmonized data sources is an extremely valuable resource for target discovery and validation efforts in the industry setting. Information about a disease-relevant target is heterogeneous and often spread across multiple databases, publications, and supplemental materials. The complex landscape of data around a target is typically best captured through a multiview approach (97, 98). For example, the data related to a target can include open, proprietary, and private data. Open Targets, with its various delivery modalities, including software, data, and projects, provides end-to-end capabilities for understanding the diverse aspects of a target and enables extensive content

customization, which brings the three streams of data classes into a unified view to offer a considerably broader perspective on a target. As an example, Sanofi's Precision Medicine & Computational Biology function has internalized both Open Targets Genetics and Open Targets Platform, incorporating them into end-to-end drug discovery workflows. This includes integrating novel internal algorithms and creating custom views designed using internal datasets. While there are other tools, both open source and proprietary, with similar functionality, the Open Targets knowledge feedback loop is unique in that insights from the data directly impact further data generation efforts whose results feed back into the Open Targets Platform. This knowledge loop creates an ever-evolving ecosystem of information that improves and refines industry partners' ability not only to link targets to diseases but also to decipher the mechanisms by which specific perturbations contribute to disease etiologies.

Another impactful use for the Open Targets Platform is the rapid triage of a list of genes, proteins, diseases, or pathways under a specific therapeutic hypothesis (**Figure 2**). An example of this is a case where a partner has a list of hundreds or even thousands of genes that are likely tractable because of their biochemical properties, predicted protein structure, or existing proprietary chemical matter. From this list, the goal is to identify a smaller set of genes causally linked to human disease traits via genetic or experimental perturbation. The Open Targets Platform allows users to efficiently prioritize targets for a deeper informatics dive and/or appropriate experimental follow-up. Notably, the data in Open Targets also help to build these deep dives (directionality via genetics, safety, animal model findings, existing drugs, supporting literature) and experimental study designs (relevant cell types, disease relevant readouts). The approach for Open Targets Genetics and Open Targets Platform has been to ensure the open-source nature of both the data and the software through a CC0 Creative Commons license and to develop the tools and means for users with different skill sets and requirements both across our partners and within the scientific community to gain access. For example, simple searches for variants, genes, traits, or drugs of interest can be conducted through an intuitive user interface, and users with more complex use cases can query through the API or Google BigQuery or download the data.

CHALLENGES AND THE FUTURE

The mission of Open Targets is to identify and prioritize therapeutic targets that will translate into safe and effective medicines. Central to this mission is the use of human genetics and genomics to generate robust mechanistic hypotheses about the drivers of human disease. In the future, therefore, much of Open Targets' activity will continue, as now, to focus on teasing out the causal threads that connect the natural experiments of genetic perturbation provided by human genetics to the pharmaceutical perturbations that enable treatments, preventions, and cures.

To date, almost all analyses of natural experiments of genetic perturbation have concentrated on disease risk, generally through the comparison of genotype frequencies between cases and controls. Yet most therapeutics for the treatment of common, late-onset diseases are given to those with established disease: Rather than aiming to halt the processes that lead to the initiation of disease, the clinical goal is typically (where full restoration of the premorbid state is not feasible) to arrest or slow progression. Until recently, the paucity of data on the genetics of disease progression had sustained the assumption that the genetics of risk and progression are sufficiently similar for the former to guide therapeutic strategies for the latter. Indeed, the evidence that targets enriched for the genetics of risk are more likely to result in safe and effective medicines provides reassurance in this regard (6, 24, 85). However, as more longitudinal datasets have become available, examples are emerging of marked divergence in the genetics of risk and progression (99). To address this divergence, Open Targets will increasingly need to be able to define and model disease as a process

through time. By doing so, it will support the community's need to adapt to this emerging understanding of the overlap and differences between the processes involved in disease initiation, those involved in progression, and therapeutic response, and use this knowledge to optimize the design of effective therapeutic strategies. As AI/ML approaches become increasingly available to explore predictive genetic and molecular biomarkers of disease risk and clinical outcome, this is a pivotal and timely opportunity to expand the acquisition of AI-ready data from clinical cohorts, enabling the concurrent collection of longitudinal molecular data encompassing genetics, cell-specific multiomics with corresponding clinical information including symptoms, diagnostic laboratory tests, treatments, response, and more.

That most of the variants implicated by human genetic analysis of common complex diseases map to regulatory sequence is now widely appreciated, but the challenge of connecting those variants to their effector genes and pathophysiological mechanisms through which they exert their impact on disease (risk or progression) remains. Open Targets will focus on addressing this using four complementary approaches.

First, the rapid expansion of the corpus of genetic data, fueled by the interrogation of datasets that are larger, more diverse, and feature richer clinical information and a more complete inventory of genetic variation, will continue to add layers of human genetic evidence. By integrating these data, Open Targets will refine appreciation of the relationship between genetic variation and phenotype, improving fine-mapping, revealing coding alleles that decipher effector gene identification, and generating allelic series that provide confidence in effect direction and titration of effect magnitude (100). The shift to sequence-based discovery improves access to rare variants, some of them annotated as having a major impact on gene function or expression. These high-impact variants also open up a path to representing impactful negative evidence within Open Targets (101). An obvious example of such evidence of the absence of genetic effect would be the failure, despite adequate power, to detect an anticipated phenotypic signal attributable to a loss-of-function allele in a putative therapeutic target, an observation that casts serious doubt on the credibility of that target.

Second, the path to mechanistic inference around genetic associations (essentially efforts to solve the variant-to-gene-to-function challenge) will increasingly run through integration with rich multiomics datasets generated from healthy and diseased tissues (including organoids) and from engineered models (most obviously, iPSC-derived cell lines). The increasing scale and range of single-cell datasets and the development of high-resolution spatial profiling methods will increasingly power insights into the specific cell types and cell states through which associated genetic variants (collectively and individually) act. This information represents an essential component to the generation of robust mechanistic hypotheses and ensures that downstream functional work is performed in the relevant cellular systems and models. Open Targets is well-placed to act as a central resource for the integration of these data, building on existing methods within Open Targets Genetics that power current L2G approaches. To date, most of this integration has been based on the overlay of genomic annotations onto genetic association datasets, but these indirect methods are increasingly being augmented by experimental approaches that provide direct examination of the impact of genetic variation on single-cell phenotypes. Examples include large-scale single cell-eQTL generation in peripheral blood mononuclear cells (44) and the development of so-called village-in-a-dish strategies based on pooled iPSC differentiation approaches (102).

Third, while the integration of genetics and genomics is powerful, it generally offers correlative, rather than causal, inference about mechanism. For example, the accuracy of inference based on the colocalization of a GWAS variant and a *cis*-eQTL signal is limited by rampant molecular pleiotropy (103, 104) and by the effects of the cell state. Perturbation studies in appropriate cellular

and model systems are now seen as an essential step in establishing the causal thread from variant to gene to disease, but this only works when they capture phenotypes that are relevant to the disease in question or, alternatively, collect high-content molecular or image-based data that can be mined for pertinent signatures of the disease state. Open Targets is already undertaking several perturbation screens in disease-relevant cellular systems, but the time is right to invest for the project to ensure that these data, together with the growing corpus of perturbation data being generated in the global research community, can be collated and cataloged in support of robust causal inference.

Ultimately the strongest mechanistic and therapeutic hypotheses will emerge when the evidence from human genetics, molecular and spatial profiling, and perturbation studies aligns in establishing the causal contribution of a given target or pathway to the disease of interest. The core competences of EMBL-EBI and Sanger ensure that Open Targets is well positioned to build inference from these diverse data types as they increase in scale, diversity, and complexity. Crucially, rather than tackling the variant-to-gene-to-disease challenge one variant at a time, inferential impact is most likely to emerge from approaches that model the biology of health and disease on a genome-wide and systematic scale. Open Targets is already heavily invested in the development of gene regulatory and protein interaction networks and is increasingly focused on the opportunities provided by more flexible approaches to data integration and data-driven inference, such as those provided by knowledge graphs and large language models. The architecture of Open Targets' open-source platforms supports the expansion of data types, visuals, and tooling to allow for the development of layering of multiple biological systems to provide users with the ability to explore and discover new insights across this wealth of data. In addition, the corpus of external and internal data accessible to Open Targets, ranging from the text of published papers and preprints, population-scale genetics and real-world data from biobanks, and atlases of single-cell expression in health and disease to genome-scale perturbation and beyond, opens the door to efforts to generate rich foundational models that, over time, will evolve into a systematic, robust, and predictive description of the biology of health and disease.

As we learn more about disease pathophysiology, the number of processes causally contributing to the risk and/or progression of any given disease (especially multifactorial disease) is set to grow, and with it the number of putative targets. Here, we define putative targets as any gene that has been mapped into one of those processes and for which it is reasonable to assume that perturbation of function or expression could have a material effect on disease. The time and effort required to undertake detailed follow-up of each such putative target place a massive premium on careful target prioritization. In part, that prioritization will be driven by the strength of the causal biology, but it also needs to take account of other factors that contribute to the likelihood that a given target can be translated into a safe and effective medicine.

These include factors that relate to safety: What is the evidence, from human genetics, from perturbation studies in cellular systems and preclinical models, from tissue expression profiles, from biological knowledge, and from experience with related drugs, that the target can be safely perturbed by therapeutics? They include factors related to druggability and tractability: How likely is it that the target will be amenable to perturbation with small molecules, antibodies, or the growing panoply of available therapeutic modalities? How many of the tools (chemical probes, assays, cellular and animal models) required to advance a drug discovery program are already in place? They also include factors related to biomarker availability: How much do we know about the downstream biology that would make it possible to develop the biomarkers of target engagement or clinical benefit that are essential elements for drug development?

Above and beyond these target-related considerations, there are questions concerning the clinical impact of a given drug development program that influence decision-making. How novel is the therapeutic hypothesis? How many other programs are targeting the same mechanism? What

is the unmet clinical need in the patient population? To what extent could stratification of the patient population (e.g., into disease subtypes or on the basis of polygenic risk) expose subsets for whom the proposed therapeutic could be particularly valuable?

The ongoing challenge is to make possible the systematic and comprehensive integration of these disparate data types to inform decisions about the targets, the pathways, the indications, and the patient populations that will maximize therapeutic success and clinical benefit. One obvious opportunity involves target prioritization within a causal pathway of interest. In situations where a target first revealed through human genetics proves problematic because of safety concerns arising from broad expression and/or tractability challenges, the systematic exploration of other targets in the biological vicinity of that original target may lead to alternative therapeutic strategies with a higher probability of success. Open Targets is uniquely positioned to assemble data that capture all of these dimensions, and more, and thereby to support pharmaceutical companies in making more informed and successful decisions about the targets they pursue.

DISCLOSURE STATEMENT

E.M. is an employee of Open Targets/EMBL-EBI, an affiliate worker for the Sanger Wellcome Institute, a member of steering committee for UK Pharmacogenetics and Stratified Medicine Network, and member of the scientific advisory committee for ELIXIR UK. G.T. is funded in part by Open Targets (grants OTAR2063, OTAR2064, OTAR2069, OTAR2076, OTAR2086), and has received financial compensation for consulting for GSK, Pfizer, Genentech, and Relation TX. M.M. is an employee of Genentech, and a holder of Roche stock. E.R.H. is an employee and stockholder at Bristol Myers Squibb. S.K. is an employee of Sanofi, a previous employee of AstraZeneca, and owns Sanofi and AstraZeneca stock. N.N. is an employee of GlaxoSmithKline. X.H. is an employee of Pfizer Inc. H.C. is an employee of Open Targets/EMBL-EBI. D.H. is an employee of Open Targets/EMBL-EBI and an affiliate worker for the Sanger Wellcome Institute.

ACKNOWLEDGMENTS

We would like to thank the Open Targets Core Team and all partners and collaborators who have contributed to the Open Targets consortium past and present. We would like to thank Irene Lopez and Yakov Tsepilov for their critical reading of the manuscript. G.T. is cofunded by Open Targets and Wellcome grant 220540/Z/20/A.

LITERATURE CITED

1. DiMasi JA, Wilkinson M. 2020. The financial benefits of faster development times: integrated formulation development, real-time manufacturing, and clinical testing. *Ther. Innov. Regul. Sci.* 54(6):1453–60
2. Hay M, Thomas DW, Craighead JL, Economides C, Rosenthal J. 2014. Clinical development success rates for investigational drugs. *Nat. Biotechnol.* 32(1):40–51
3. Deloitte Cent. Health Solut. 2022. *Nurturing growth: measuring the return from pharmaceutical innovation 2021*. Rep., Deloitte Cent. Health Solut., London. <https://www2.deloitte.com/content/dam/Deloitte/uk/Documents/life-sciences-health-care/Measuring-the-return-of-pharmaceutical-innovation-2021-Deloitte.pdf>
4. Ochoa D, Hercules A, Carmona M, Suveges D, Gonzalez-Urriarte A, et al. 2021. Open Targets Platform: supporting systematic drug–target identification and prioritisation. *Nucleic Acids Res.* 49(D1):D1302–10
5. Ghoussaini M, Mountjoy E, Carmona M, Peat G, Schmidt EM, et al. 2021. Open Targets Genetics: systematic identification of trait-associated genes using large-scale genetics and functional genomics. *Nucleic Acids Res.* 49(D1):D1311–20

6. Nelson MR, Tipney H, Painter JL, Shen J, Nicoletti P, et al. 2015. The support of human genetic evidence for approved drug indications. *Nat. Genet.* 47(8):856–60
7. Ochoa D, Karim M, Ghoussaini M, Hulcoop DG, McDonagh EM, Dunham I. 2022. Human genetics evidence supports two-thirds of the 2021 FDA-approved drugs. *Nat. Rev. Drug Discov.* 21(8):551
8. Rusina PV, Falaguera MJ, Romero JMR, McDonagh EM, Dunham I, Ochoa D. 2023. Genetic support for FDA-approved drugs over the past decade. *Nat. Rev. Drug Discov.* 22(11):864
9. Buniello A, MacArthur JAL, Cerezo M, Harris LW, Hayhurst J, et al. 2019. The NHGRI-EBI GWAS Catalog of published genome-wide association studies, targeted arrays and summary statistics 2019. *Nucleic Acids Res.* 47(D1):D1005–12
10. Kerimov N, Hayhurst JD, Peikova K, Manning JR, Walter P, et al. 2021. A compendium of uniformly processed human gene expression and splicing quantitative trait loci. *Nat. Genet.* 53(9):1290–99
11. Soskic B, Cano-Gamez E, Smyth DJ, Ambridge K, Ke Z, et al. 2022. Immune disease risk variants regulate gene expression dynamics during CD4⁺ T cell activation. *Nat. Genet.* 54(6):817–26
12. Schwartzentruber J, Cooper S, Liu JZ, Barrio-Hernandez I, Bello E, et al. 2021. Genome-wide meta-analysis, fine-mapping and integrative prioritization implicate new Alzheimer's disease risk genes. *Nat. Genet.* 53(3):392–402
13. Barrio-Hernandez I, Schwartzentruber J, Shrivastava A, Del-Toro N, Gonzalez A, et al. 2023. Network expansion of genetic associations defines a pleiotropy map of human cell biology. *Nat. Genet.* 55(3):389–98
14. Mountjoy E, Schmidt EM, Carmona M, Schwartzentruber J, Peat G, et al. 2021. An open approach to systematically prioritize causal variants and genes at all published human GWAS trait-associated loci. *Nat. Genet.* 53(11):1527–33
15. Krzak M, Alegbe T, Taylor DL, Ghouraba M, Strickland M, et al. 2023. Single-cell RNA sequencing reveals dysregulated cellular programmes in the inflamed epithelium of Crohn's disease patients. medRxiv 2023.09.06.23295056. <https://doi.org/10.1101/2023.09.06.23295056>
16. Huang H, Fang M, Jostins L, Umićević Mirkov M, Boucher G, et al. 2017. Fine-mapping inflammatory bowel disease loci to single-variant resolution. *Nature* 547(7662):173–78
17. Cooper SE, Schwartzentruber J, Bello E, Coomber EL, Bassett AR. 2020. Screening for functional transcriptional and splicing regulatory variants with GenIE. *Nucleic Acids Res.* 48(22):e131
18. Trynka G, Sandor C, Han B, Xu H, Stranger BE, et al. 2013. Chromatin marks identify critical cell types for fine mapping complex trait variants. *Nat. Genet.* 45(2):124–30
19. Soskic B, Cano-Gamez E, Smyth DJ, Rowan WC, Nakic N, et al. 2019. Chromatin activity at GWAS loci identifies T cell states driving complex immune diseases. *Nat. Genet.* 51(10):1486–93
20. Behan FM, Iorio F, Picco G, Gonçalves E, Beaver CM, et al. 2019. Prioritization of cancer therapeutic targets using CRISPR-Cas9 screens. *Nature* 568(7753):511–16
21. Chan EM, Shibue T, McFarland JM, Gaeta B, Ghandi M, et al. 2019. WRN helicase is a synthetic lethal target in microsatellite unstable cancers. *Nature* 568(7753):551–56
22. Pacini C, Dempster JM, Boyle I, Gonçalves E, Najgebauer H, et al. 2021. Integrated cross-study datasets of genetic dependencies in cancer. *Nat. Commun.* 12(1):1661
23. Dwane L, Behan FM, Gonçalves E, Lightfoot H, Yang W, et al. 2021. Project Score database: a resource for investigating cancer cell dependencies and prioritizing therapeutic targets. *Nucleic Acids Res.* 49(D1):D1365–72
24. King EA, Davis JW, Degner JF. 2019. Are drug targets with genetic support twice as likely to be approved? Revised estimates of the impact of genetic support for drug mechanisms on the probability of drug approval. *PLOS Genet.* 15(12):e1008489
25. Zhou W, Nielsen JB, Fritsche LG, Dey R, Gabrielsen ME, et al. 2018. Efficiently controlling for case-control imbalance and sample relatedness in large-scale genetic association studies. *Nat. Genet.* 50(9):1335–41
26. Malone J, Holloway E, Adamusiak T, Kapushesky M, Zheng J, et al. 2010. Modeling sample variables with an Experimental Factor Ontology. *Bioinformatics* 26(8):1112–18
27. Sollis E, Mosaku A, Abid A, Buniello A, Cerezo M, et al. 2023. The NHGRI-EBI GWAS Catalog: knowledgebase and deposition resource. *Nucleic Acids Res.* 51(D1):D977–85

28. GTEx Consortium. 2020. The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science* 369(6509):1318–30
29. Vösa U, Claringbould A, Westra H-J, Bonder MJ, Deelen P, et al. 2018. Unraveling the polygenic architecture of complex traits using blood eQTL metaanalysis. *bioRxiv* 447367. <https://doi.org/10.1101/447367>
30. Sun BB, Maranville JC, Peters JE, Stacey D, Staley JR, et al. 2018. Genomic atlas of the human plasma proteome. *Nature* 558(7708):73–79
31. Folkersen L, Fauman E, Sabater-Lleal M, Strawbridge RJ, Frånberg M, et al. 2017. Mapping of 79 loci for 83 plasma protein biomarkers in cardiovascular disease. *PLOS Genet.* 13(4):e1006706
32. Hillary RF, McCartney DL, Harris SE, Stevenson AJ, Seebth A, et al. 2019. Genome and epigenome wide studies of neurological protein biomarkers in the Lothian Birth Cohort 1936. *Nat. Commun.* 10(1):3160
33. Ahola-Olli AV, Würtz P, Havulinna AS, Aalto K, Pitkänen N, et al. 2017. Genome-wide association study identifies 27 loci influencing concentrations of circulating cytokines and growth factors. *Am. J. Hum. Genet.* 100(1):40–50
34. Pietzner M, Stewart ID, Raffler J, Khaw K-T, Michelotti GA, et al. 2021. Plasma metabolites to profile pathways in noncommunicable disease multimorbidity. *Nat. Med.* 27(3):471–79
35. Suhre K, Arnold M, Bhagwat AM, Cotton RJ, Engelke R, et al. 2017. Connecting genetic risk to disease end points through the human blood plasma proteome. *Nat. Commun.* 8(1):14357
36. Folkersen L, Gustafsson S, Wang Q, Hansen DH, Hedman ÅK, et al. 2020. Genomic and drug target evaluation of 90 cardiovascular proteins in 30,931 individuals. *Nat. Metab.* 2(10):1135–48
37. McLaren W, Gil L, Hunt SE, Riat HS, Ritchie GRS, et al. 2016. The Ensembl Variant Effect Predictor. *Genome Biol.* 17(1):122
38. Javierre BM, Burren OS, Wilder SP, Kreuzhuber R, Hill SM, et al. 2016. Lineage-specific genome architecture links enhancers and non-coding disease variants to target gene promoters. *Cell* 167(5):1369–84.e19
39. Andersson R, Gebhard C, Miguel-Escalada I, Hoof I, Bornholdt J, et al. 2014. An atlas of active enhancers across human cell types and tissues. *Nature* 507(7493):455–61
40. Thurman RE, Rynes E, Humbert R, Vierstra J, Maurano MT, et al. 2012. The accessible chromatin landscape of the human genome. *Nature* 489(7414):75–82
41. Davies M, Nowotka M, Papadatos G, Dedman N, Gaulton A, et al. 2015. ChEMBL web services: streamlining access to drug discovery data and utilities. *Nucleic Acids Res.* 43(W1):W612–20
42. Mendez D, Gaulton A, Bento AP, Chambers J, De Veij M, et al. 2019. ChEMBL: towards direct deposition of bioassay data. *Nucleic Acids Res.* 47(D1):D930–40
43. Stacey D, Fauman EB, Ziemek D, Sun BB, Harshfield EL, et al. 2019. ProGeM: a framework for the prioritization of candidate causal genes at molecular quantitative trait loci. *Nucleic Acids Res.* 47(1):e3
44. van der Wijst M, de Vries DH, Groot HE, Trynka G, Hon CC, et al. 2020. The single-cell eQTLGen consortium. *eLife* 9:e52155
45. Martin FJ, Amode MR, Aneja A, Austine-Orimoloye O, Azov AG, et al. 2023. Ensembl 2023. *Nucleic Acids Res.* 51(D1):D933–41
46. Strande NT, Riggs ER, Buchanan AH, Ceyhan-Birsoy O, DiStefano M, et al. 2017. Evaluating the clinical validity of gene-disease associations: an evidence-based framework developed by the Clinical Genome Resource. *Am. J. Hum. Genet.* 100(6):895–906
47. Martin AR, Williams E, Foulger RE, Leigh S, Daugherty LC, et al. 2019. PanelApp crowdsources expert knowledge to establish consensus diagnostic gene panels. *Nat. Genet.* 51(11):1560–65
48. Thormann A, Halachev M, McLaren W, Moore DJ, Svinti V, et al. 2019. Flexible and scalable diagnostic filtering of genomic variants using G2P with Ensembl VEP. *Nat. Commun.* 10(1):2373
49. Maiella S, Rath A, Angin C, Mousson F, Kremp O. 2013. Orphanet et son réseau: où trouver une information validée sur les maladies rares [Orphanet and its consortium: where to find expert-validated information on rare diseases]. *Rev. Neurol.* 169(Suppl. 1):S3–8
50. DiStefano MT, Goehringer S, Babb L, Alkuraya FS, Amberger J, et al. 2022. The Gene Curation Coalition: a global effort to harmonize gene–disease evidence resources. *Genet. Med.* 24(8):1732–42

51. Wang Q, Dhindsa RS, Carss K, Harper AR, Nag A, et al. 2021. Rare variant contribution to human disease in 281,104 UK Biobank exomes. *Nature* 597(7877):527–32
52. Backman JD, Li AH, Marcketta A, Sun D, Mbatchou J, et al. 2021. Exome sequencing and analysis of 454,787 UK Biobank participants. *Nature* 599(7886):628–34
53. Karczewski KJ, Solomonson M, Chao KR, Goodrich JK, Tiao G, et al. 2022. Systematic single-variant and gene-based association testing of thousands of phenotypes in 394,841 UK Biobank exomes. *Cell Genom.* 2(9):100168
54. Zhou X, Feliciano P, Shu C, Wang T, Astrovskaia I, et al. 2022. Integrating de novo and inherited variants in 42,607 autism cases identifies mutations in new moderate-risk genes. *Nat. Genet.* 54(9):1305–19
55. Singh T, Poterba T, Curtis D, Akil H, Al Eissa M, et al. 2022. Rare coding variants in ten genes confer substantial risk for schizophrenia. *Nature* 604(7906):509–16
56. Satterstrom FK, Kosmicki JA, Wang J, Breen MS, De Rubeis S, et al. 2020. Large-scale exome sequencing study implicates both developmental and functional changes in the neurobiology of autism. *Cell* 180(3):568–84.e23
57. Epi25 Collab. 2019. Ultra-rare genetic variation in the epilepsies: a whole-exome sequencing study of 17,606 individuals. *Am. J. Hum. Genet.* 105(2):267–82
58. Bomba L, Walter K, Guo Q, Surendran P, Kundu K, et al. 2022. Whole-exome sequencing identifies rare genetic variants associated with human plasma metabolites. *Am. J. Hum. Genet.* 109(6):1038–54
59. Akbari P, Sosina OA, Bovijn J, Landheer K, Nielsen JB, et al. 2022. Multiancestry exome sequencing reveals INHBE mutations associated with favorable fat distribution and protection from diabetes. *Nat. Commun.* 13(1):4844
60. Makarios MB, Lake J, Pitz V, Ye Fu A, Guidubaldi JL, et al. 2023. Large-scale rare variant burden testing in Parkinson's disease. *Brain* 146(11):4622–32
61. Riveros-Mckay F, Oliver-Williams C, Karthikeyan S, Walter K, Kundu K, et al. 2020. The influence of rare variants in circulating metabolic biomarkers. *PLOS Genet.* 16(3):e1008605
62. Futreal PA, Coin L, Marshall M, Down T, Hubbard T, et al. 2004. A census of human cancer genes. *Nat. Rev. Cancer* 4(3):177–83
63. Tate JG, Bamford S, Jubb HC, Sondka Z, Beare DM, et al. 2019. COSMIC: the Catalogue Of Somatic Mutations In Cancer. *Nucleic Acids Res.* 47(D1):D941–47
64. Martínez-Jiménez F, Muiños F, Sentís I, Deu-Pons J, Reyes-Salazar I, et al. 2020. A compendium of mutational cancer driver genes. *Nat. Rev. Cancer* 20(10):555–72
65. Landrum MJ, Lee JM, Benson M, Brown GR, Chao C, et al. 2018. ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Res.* 46(D1):D1062–67
66. Ghousaini M, Nelson MR, Dunham I. 2023. Future prospects for human genetics and genomics in drug discovery. *Curr. Opin. Struct. Biol.* 80:102568
67. Schneider M, Radoux CJ, Hercules A, Ochoa D, Dunham I, et al. 2021. The PROTACtable genome. *Nat. Rev. Drug Discov.* 20(10):789–97
68. Jumper J, Evans R, Pritzel A, Green T, Figurnov M, et al. 2021. Highly accurate protein structure prediction with AlphaFold. *Nature* 596(7873):583–89
69. Tunyasuvunakool K, Adler J, Wu Z, Green T, Zielinski M, et al. 2021. Highly accurate protein structure prediction for the human proteome. *Nature* 596(7873):590–96
70. Roberts TC, Wood MJA, Davies KE. 2023. Therapeutic approaches for Duchenne muscular dystrophy. *Nat. Rev. Drug Discov.* 22(11):917–34
71. Müller S, Ackloo S, Al Chawaf A, Al-Lazikani B, Antolin A, et al. 2022. Target 2035 - update on the quest for a probe for every protein. *RSC Med. Chem.* 13(1):13–21
72. Bowes J, Brown AJ, Hamon J, Jarolimek W, Sridhar A, et al. 2012. Reducing safety-related drug attrition: the use of in vitro pharmacological profiling. *Nat. Rev. Drug Discov.* 11(12):909–22
73. Force T, Kolaja KL. 2011. Cardiotoxicity of kinase inhibitors: the prediction and translation of preclinical models to clinical outcomes. *Nat. Rev. Drug Discov.* 10(2):111–26
74. Richard AM, Judson RS, Houck KA, Grulke CM, Volarath P, et al. 2016. ToxCast chemical landscape: paving the road to 21st century toxicology. *Chem. Res. Toxicol.* 29(8):1225–51

75. Lamore SD, Ahlberg E, Boyer S, Lamb ML, Hortigon-Vinagre MP, et al. 2017. Deconvoluting kinase inhibitor induced cardiotoxicity. *Toxicol. Sci.* 158(1):213–26
76. Lynch JJ, Van Vleet TR, Mittelstadt SW, Blomme EAG. 2017. Potential functional and pathological side effects related to off-target pharmacological activity. *J. Pharmacol. Toxicol. Methods* 87:108–26
77. Urban L, Whitebread S, Hamon J, Mikhailov D, Azzaoui K. 2012. Screening for safety-relevant off-target activities. In *Polypharmacology in Drug Discovery*, ed. J-U Peters, pp. 15–46. Hoboken, NJ: Wiley
78. Carss KJ, Deaton AM, Del Rio-Espinola A, Diogo D, Fielden M, et al. 2023. Using human genetics to improve safety assessment of therapeutics. *Nat. Rev. Drug Discov.* 22(2):145–62
79. Chen S, Francioli LC, Goodrich JK, Collins RL, Kanai M, et al. 2022. A genome-wide mutational constraint map quantified from variation in 76,156 human genomes. *bioRxiv* 2022.03.20.485034. <https://doi.org/10.1101/2022.03.20.485034>
80. Karczewski KJ, Francioli LC, Tiao G, Cummings BB, Alföldi J, et al. 2020. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature* 581:434–43
81. Papatheodorou I, Moreno P, Manning J, Fuentes AM-P, George N, et al. 2020. Expression Atlas update: from tissues to single cells. *Nucleic Acids Res.* 48(D1):D77–83
82. Blake JA, Baldarelli R, Kadin JA, Richardson JE, Smith CL, et al. 2021. Mouse Genome Database (MGD): knowledgebase for mouse-human comparative biology. *Nucleic Acids Res.* 49(D1):D981–87
83. Maciejewski M, Lounkine E, Whitebread S, Farmer P, DuMouchel W, et al. 2017. Reverse translation of adverse event reports paves the way for de-risking preclinical off-targets. *eLife* 6:e25818
84. Ochoa D, Hercules A, Carmona M, Suveges D, Baker J, et al. 2023. The next-generation Open Targets Platform: reimaged, redesigned, rebuilt. *Nucleic Acids Res.* 51(D1):D1353–59
85. Razuvayevskaya O, Lopez I, Dunham I, Ochoa D. 2023. Why clinical trials stop: the role of genetics. *medRxiv* 2023.02.07.23285407. <https://doi.org/10.1101/2023.02.07.23285407>
86. Han Y, Klinger K, Rajpal DK, Zhu C, Teeple E. 2022. Empowering the discovery of novel target-disease associations via machine learning approaches in the Open Targets platform. *BMC Bioinform.* 23(1):232
87. Gogleva A, Polychronopoulos D, Pfeifer M, Poroshin V, Ughetto M, et al. 2022. Knowledge graph-based recommendation framework identifies drivers of resistance in EGFR mutant non-small cell lung cancer. *Nat. Commun.* 13(1):1667
88. Ye C, Swiers R, Bonner S, Barrett I. 2022. A knowledge graph-enhanced tensor factorisation model for discovering drug targets. *IEEE/ACM Trans. Comput. Biol. Bioinform.* 19(6):3070–80
89. Fernández-Torras A, Duran-Frigola M, Bertoni M, Locatelli M, Aloy P. 2022. Integrating and formatting biomedical data as pre-calculated knowledge graph embeddings in the Bioteque. *Nat. Commun.* 13(1):5304
90. Geleta D, Nikolov A, Edwards G, Gogleva A, Jackson R, et al. 2021. Biological Insights Knowledge Graph: an integrated knowledge graph to support drug development. *bioRxiv* 2021.10.28.466262. <https://doi.org/10.1101/2021.10.28.466262>
91. Paliwal S, de Giorgio A, Neil D, Michel J-B, Lacoste AM. 2020. Preclinical validation of therapeutic targets predicted by tensor factorization on heterogeneous graphs. *Sci. Rep.* 10(1):18250
92. Failli M, Paananen J, Fortino V. 2019. Prioritizing target-disease associations with novel safety and efficacy scoring methods. *Sci. Rep.* 9(1):9852
93. Lee C, Lin J, Prokop A, Gopalakrishnan V, Hanna RN, et al. 2022. StarGazer: a hybrid intelligence platform for drug target prioritization and digital drug repositioning using Streamlit. *Front. Genet.* 13:868015
94. Cezard T, Cunningham F, Hunt SE, Koyle B, Kumar N, et al. 2022. The European Variation Archive: a FAIR resource of genomic variation for all species. *Nucleic Acids Res.* 50(D1):D1216–20
95. Moreno P, Fexova S, George N, Manning JR, Miao Z, et al. 2022. Expression Atlas update: gene and protein expression in multiple species. *Nucleic Acids Res.* 50(D1):D129–40
96. Varusai TM, Jupe S, Sevilla C, Matthews L, Gillespie M, et al. 2021. Using Reactome to build an autophagy mechanism knowledgebase. *Autophagy* 17(6):1543–54
97. Shameer K, Badgeley MA, Miotto R, Glicksberg BS, Morgan JW, Dudley JT. 2017. Translational bioinformatics in the era of real-time biomedical, health care and wellness data streams. *Brief. Bioinform.* 18(1):105–24

98. Weissler EH, Naumann T, Andersson T, Ranganath R, Elemento O, et al. 2021. The role of machine learning in clinical research: transforming the future of evidence generation. *Trials* 22(1):537
99. Int. Mult. Scler. Genet. Consort., MultipleMS Consort. 2023. Locus for severity implicates CNS resilience in progression of multiple sclerosis. *Nature* 619(7969):323–31
100. Dendrou CA, Cortes A, Shipman L, Evans HG, Attfield KE, et al. 2016. Resolving TYK2 locus genotype-to-phenotype differences in autoimmunity. *Sci. Transl. Med.* 8(363):363ra149
101. McGregor TL, Hunt KA, Yee E, Mason D, Nioi P, et al. 2020. Characterising a healthy adult with a rare HAO1 knockout to support a therapeutic strategy for primary hyperoxaluria. *eLife* 9:e54363
102. Jerber J, Seaton DD, Cuomo ASE, Kumasaka N, Haldane J, et al. 2021. Population-scale single-cell RNA-seq profiling across dopaminergic neuron differentiation. *Nat. Genet.* 53(3):304–12
103. Tambets R, Kolde A, Kolberg P, Love MI, Alasoo K. 2023. Extensive co-regulation of neighbouring genes complicates the use of eQTLs in target gene prioritization. bioRxiv 2023.09.29.560109. <https://doi.org/10.1101/2023.09.29.560109>
104. Ndungu A, Payne A, Torres JM, van de Bunt M, McCarthy MI. 2020. A multi-tissue transcriptome analysis of human metabolites guides interpretability of associations based on multi-SNP models for gene expression. *Am. J. Hum. Genet.* 106(2):188–201